

closure_evidence_20260923

BOB-201 closure evidence — 2026-09-23

Operator decision implemented

“Resolve symlinks (realpath) and re-fence” (option a).
ownership_path_fence() (scripts/lib/ownership.sh, the single shared implementation used by both scripts/ownership_repair.sh and scripts/ownership_precondition.sh) now resolves every intermediate path component that is a symlink via a real filesystem check (readlink -f per-component, with [[-e "\$target"]] existence verification closing a subtlety plain rc-code checking misses: a standalone dangling symlink returns rc=0 with its raw nonexistent target text) BEFORE the existing lexical containment/denylist checks judge the path. The FINAL (leaf) path component is deliberately never resolved (already covered by find’s own -P behaviour per the suite’s existing Case 17), and a genuinely-not-yet-created component (never a symlink) still falls through to the pre-fix lexical-append behaviour, preserving the legitimate “declared download root not yet created” case (data-model E1) — this data-loss-avoidance edge case was explicitly identified and protected against during implementation, not merely assumed safe.

Independently re-verified this session (coordinator, including a genuine

paired-mutation revert/restore reproduced from a clean shell)

```
$ bash tests/unit/test_ownership_repair.sh
... Case 25: an intermediate symlink is resolved before the fence
judges the path (BOB-201)
    PASS x7 (all 7 sub-cases, including the golden-FALSE symlinked-
```

but-in-scope

case and both not-yet-created-path non-regression guards)

RESULT: 153 passed, 0 failed, 0 skipped

```
$ git stash push -m "coordinator-verify-bob201-mutation" --  
scripts/lib/ownership.sh
```

```
$ bash tests/unit/test_ownership_repair.sh
```

FAIL x4 (the four symlink-escape variants – genuinely reproduces the

pre-fix defect) + Case 24's own citation-text assertion also correctly

fails against the reverted file (an expected 5th failure, not a bug)

RESULT: 148 passed, 5 failed, 0 skipped

```
$ git stash pop
```

```
$ bash tests/unit/test_ownership_repair.sh
```

RESULT: 153 passed, 0 failed, 0 skipped

```
$ bash tests/unit/test_ownership_precondition.sh
```

RESULT: 29 passed, 0 failed, 0 skipped

```
$ bash tests/unit/test_ownership_rootless_detection.sh
```

RESULT: 19 passed, 0 failed

Both sibling suites (which share the fixed function) unaffected.

```
$ grep -n "judges the spelling\|JUDGES THE SPELLING" scripts/lib/  
ownership.sh tests/unit/test_ownership_repair.sh  
(empty)
```

The stale “IT JUDGES THE SPELLING; THE KERNEL RESOLVES THE PATH” claim (now false — intermediate components genuinely ARE resolved) was corrected in-source alongside Case 24’s matching assertion, per §11.4.6 (a claim must match measured reality). All existing BOB-159 citations were audited (grep -rn “BOB-159” scripts/ tests/) and confirmed to be deliberate historical narrative already correctly resolved to BOB-201 by an earlier commit (03bf860) — none were live/incorrect, none were touched.

Honest boundary (not silenced)

TOCTOU is accepted, not eliminated: the fence resolves at judgment time; the later recursive walk uses the original (unresolved) declared path. A symlink swapped in between fence-time and walk-time is not caught by this fix — the accepted cost of the operator’s chosen resolve-and-reference direction, documented explicitly in-source rather than implying stronger guarantees than are actually provided. A

permission-denied plain (non-symlink) intermediate component is treated as “not yet created” (accepted, falls through), not refused — judged in-scope-correct since the original lexical fence never checked filesystem permissions at all.

git diff –stat

```
scripts/lib/ownership.sh          | 206 ++++++
+++++---
tests/unit/test_ownership_repair.sh | 177 ++++++
++++--
2 files changed, 346 insertions(+), 37 deletions(-)
```